

Part 2 Risk**CHAPTER****9**

Risk and the Cost of Capital

Long before the development of modern theories linking risk and return, smart financial managers adjusted for risk in capital budgeting. They knew that risky projects are, other things equal, less valuable than safe ones—that's just common sense. Therefore, they demanded higher rates of return from risky projects, or they based their decisions about risky projects on conservative forecasts of project cash flows.

In this chapter, we consider how to incorporate risk into investment decisions. There are a number of pitfalls in including risk in capital budgeting analyses that are important to understand and avoid.

Section 9.1 Company and project costs of capital

Today, most companies start with the company cost of capital as a benchmark discount rate for new investments. The company cost of capital is the right discount rate only for investments that have the same risk as the company's overall business. For riskier projects, the cost of capital is greater than the company cost of capital. For safer projects, it is less.

One way to estimate the company cost of capital is to estimate the company's asset beta, which is a weighted average of its debt and equity betas. The asset beta can then be plugged into the capital asset pricing model to derive the company cost of capital. A more convenient alternative is to calculate a weighted average of the returns demanded by investors in the company's debt and equity.

Section 9.2 Estimating beta and the cost of capital

The hardest part of estimating the weighted-average cost of capital is figuring out the cost of equity—that is, the expected rate of return to investors in the firm's common stock. Many firms turn to the capital asset pricing model (CAPM) for an answer. The CAPM states that the expected rate of return equals the risk-free interest rate plus a risk premium that depends on beta and the market risk premium.

You can look up betas on financial websites like Yahoo! Finance and Bloomberg, but it's important to remember that these betas are estimates and liable to statistical errors. We look at how to estimate betas and to check the reliability of these estimates.

Section 9.3 Analyzing project risk

Suppose you're responsible for a specific investment project. How do you know if the project is average risk or

above- or below-average risk? We suggest you to check whether the project's cash flows are more or less sensitive to the business cycle than the average project. Also check whether the project has higher or lower fixed operating costs (higher or lower operating leverage) and whether it requires large future investments.

Remember that a project's cost of capital depends only on market risk. Diversifiable events can affect project cash flows but they do not increase the cost of capital. Also don't be tempted to add arbitrary fudge factors to discount rates to try to adjust for risk. A fudge factor is a figure included in a calculation to account for an unquantified influence. They are too often added to discount rates for projects in unstable parts of the world, for example.

Section 9.4 Certainty equivalents

Financial managers usually use a risk-adjusted discount rate for all future cash flows. An alternative is to estimate certainty-equivalent cash flows that can then be discounted at the risk-free interest rate. Although rarely used for standard DCF calculations, certainty equivalents are vital for valuing options and futures. We also use them to illustrate the deductions for risk that are assumed when we use a single risk-adjusted discount rate.

9-1 Company and Project Costs of Capital

The **company cost of capital** is the opportunity cost of capital for an investment in all the firm's assets and is the correct discount rate for its average-risk projects.

If the company has no debt outstanding, the company cost of capital equals its cost of equity, the expected rate of return on the company's common stock. A few large, successful companies pretty well fit this no-debt special case, including Johnson & Johnson (J&J).¹ Its market-value balance sheet at the end of the third quarter of 2020 boiled down to:

J&J Market Value, billions			
Assets: A = \$380	$\beta_A = 0.75$	Equity: E = \$380	$\beta_E = 0.75$

The equity value of \$380 billion was a market capitalization, not a book (accounting) value. The value of assets and equity on J&J's book balance sheet was only about \$60 billion. We can't observe the market value of J&J's assets on its book financial statements, but we can infer it from the market cap.

We estimated J&J's equity beta in Table 8.1 as 0.75. Therefore its asset beta must also be $\beta_A = 0.75$. J&J's shareholders bear all the risks of J&J's assets. Beta must be the same on

both sides of its balance sheet, in this case $\beta_E = \beta_A = 0.75$.

We can use the CAPM to calculate J&J's company cost of capital r_A . If the risk-free interest rate is 2% and the market risk premium is 7%,

$$r_A = r_f + \beta_A (r_m - r_f) = 2 + 0.75 \times 7 = 7.3\%$$

If J&J considers an average-risk investment, it makes sense to discount the forecasted cash flows at 7.3%.

Company Cost of Capital for CSX

Now let's try the same exercise for the railroad CSX, which has both debt and equity outstanding. Its balance sheet at the end of the third quarter of 2020 included \$16.5 billion of debt. The market value of CSX's equity was \$60.0 billion, well above the \$12.7 billion book equity value. So its debt-to-value ratio was 22%, much lower than a debt ratio calculated from its financial statements.

CSX Market Value, billions			
Assets: A = \$76.5	$\beta_A = ?$	Debt: D = \$16.5 (22%)	$\beta_D = 0.2$
		Equity: E = \$60 (78%)	$\beta_E = 1.18$
Total: V = \$76.5		Total: V = \$76.5	

What was CSX's asset beta? Not the debt beta because debt has a senior claim on the assets and is therefore safer than the assets. Not the equity beta because debt amplifies shareholders' risk. Shareholders bear both business and *financial risk* because borrowing increases the volatility and beta of equity.

We can't reach the asset beta by looking at debt or equity betas separately. We need a *portfolio* of all of CSX's debt and equity financing. Suppose you purchased a portfolio of 100% of the debt and 100% of the equity. Then you would own 100% of the assets, and the portfolio beta would equal the asset beta. The weights of debt and equity in such a portfolio would be D/V and E/V calculated from *market* values. (Investors can't buy CSX shares for book value.)

CSX's equity beta was $\beta_E = 1.18$, and we will assume a debt beta of $\beta_D = 0.2$. The asset beta is:

$$\beta_A = \beta_D D/V + \beta_E E/V = 0.2 \times 0.22 + 1.18 \times 0.78 = 0.96$$

The company cost of capital for CSX, using the same risk-free rate and market risk premium as for J&J, is:

$$r_A = r_f + \beta_A (r_m - r_f) = 2 + 0.96 \times 7 = 8.8, \text{ about } 9\%$$

9.1 Self-Test

Suppose you mistakenly use a debt ratio from book financial statements instead of market values. Are your estimates of the company cost of capital likely to be too high or too low? Assume you are analyzing a successful public company.

A Weighted Average of Betas, or a Weighted Average of Returns? We have introduced a fundamental relationship: Risk must be the same on both sides of the balance sheet. If risk means beta, the formula is:

$$\text{Asset beta} = \beta_A = \beta_D D/V + \beta_E E/V \quad (9.1)$$

We used the asset beta β_A to calculate the company cost of capital for J&J and CSX.

Yet we have skipped over an important practical problem. We assumed a debt beta. How does a financial manager know what the debt beta is?² She won't find debt betas quoted in *The Wall Street Journal*, Yahoo! Finance, or other standard sources of financial data.

Most financial managers avoid this problem. Why go to the trouble of estimating debt betas when the cost of debt appears in plain sight? It is the interest rate paid on corporate borrowing. CSX's CFO knew in 2020 what the interest rate would be on a new bond issue. The CFO did not need a debt beta to estimate that interest rate.

A CFO is usually better off averaging the *costs* of debt and equity, not their betas. We can still use the CAPM to calculate the cost of equity from CSX's equity beta of 1.18:

$$r_E = r_f + \beta_E (r_m - r_f) = 2 + 1.18 \times 7 = 10.3\%$$

If the cost of debt is $r_D = 3.4\%$, the company cost of capital is:

$$r_A = r_D D/V + r_E E/V = 3.4 \times 0.22 + 10.3 \times 0.78 = 8.8, \text{ about } 9\%$$

which in this case is identical to the cost of capital calculated using the asset beta:³

We have a second fundamental relationship: The expected rate of return is the same on both sides of the balance sheet:

$$\text{Company cost of capital} = r_A = r_D D/V + r_E E/V \quad (9.2)$$

The two relationships in Eqs. (9.1) and (9.2) are both important, and we will use both in various settings later in the book. Equation (9.2), which averages expected returns rather than betas, is more common in practice for estimating the company cost of capital. It has two practical advantages. First, estimating the cost of debt for Eq. (9.2) does not require a debt beta. The equation uses the actual cost of borrowing. Second, although the CAPM is widely used to estimate the cost of equity, it is not required in Eq. (9.2). The financial manager may decide to use a DCF model from Chapter 4, for example.

9.2 Self-Test

Viking Runeship's book balance sheet reports 50% debt and 50% equity. The debt is trading close to book value at a current interest rate of 4%. The market cap of Runeship's equity is double book value. The equity beta is $\beta_E = 1$. The risk-free interest rate is 2%, and the market risk premium is 7%. What is Runeship's company cost of capital?

Three Warnings

Is debt cheap finance? The formula for the company cost of capital blends the costs of debt and equity. Therefore, you might think that the average cost of capital could be reduced by substituting cheap debt for expensive equity. For example, you might expect that CSX could reduce its cost of capital by substituting debt costing only 3.4% for equity which costs more than 10%.

It doesn't work that way! As the debt ratio D/V increases, the cost of the remaining equity also increases, offsetting the apparent advantage of more cheap debt. We show how and why this offset happens in Chapter 16.

What is the risk-free rate? The next issue is what value to use for the risk-free interest rate. Interest rates at the end of 2020 were at a record low—much lower than the 2% rate used in the J&J and CSX examples. The three-month Treasury bill rate was barely positive at 0.1%. Yields on 20-year U.S. Treasury bonds were about 1.4%.

The CAPM is a short-term model. It works period by period and calls for a short-term interest rate. But could a 0.1% three-month risk-free rate give the right discount rate for cash flows 10 or 20 years in the future? Well, now that you mention it, probably not.

Financial managers have to muddle through this problem. Most use a long-term risk-free rate in the CAPM formula.⁴ If this short-cut is used, then the market risk premium must be restated as the difference between market returns and returns on *long-term* Treasuries.⁵ Table 7.1 shows that over the past century, this risk premium has averaged 6.2%.

Interest is tax deductible. Equation (9.2) is a weighted-average cost of capital (WACC). But it is not the same thing as the *after-tax* weighted-average cost of capital (*after-tax* WACC), which uses the *after-tax* cost of debt to capture the value of taxes saved by deducting interest from taxable income. Many companies use the after-tax WACC as a standard discount rate for average-risk investment projects.

We defer discussion of the after-tax WACC to Chapter 18, after we have covered the pros and cons of debt financing in detail. That chapter also covers the adjusted present value (APV) valuation method, in which tax savings from interest deductions are forecasted and valued separately. The company cost of capital is the correct APV discount rate for average-

risk investments. In this chapter, we defer consideration of interest tax deductions and keep the focus on the company cost of capital.

What about Investments That Are Not Average Risk?

The company cost of capital is *not* the correct discount rate if a new investment project is more or less risky than the firm's existing business. Each project should, in principle, be evaluated at its *own* opportunity cost of capital. This is a clear implication of the value-additivity principle introduced in Chapter 5. For a firm composed of assets A and B, the firm value is

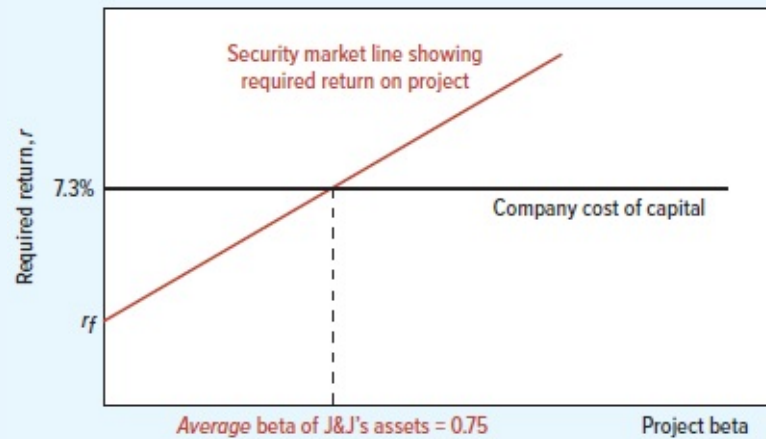
$$\begin{aligned}\text{Firm value} &= \text{PV}(AB) = \text{PV}(A) + \text{PV}(B) \\ &= \text{sum of separate asset values}\end{aligned}$$

Here, $\text{PV}(A)$ and $\text{PV}(B)$ are valued just as if they were mini-firms that stockholders could invest in directly. Investors would value A by discounting its forecasted cash flows at a rate reflecting the risk of A. They would value B by discounting at a rate reflecting the risk of B. The two discount rates will, in general, be different. If the present value of an asset depended on the identity of the company that bought it, present values would *not* add up, and we know they do add up. (Think of a portfolio of \$1 million invested in J&J and \$1 million invested in CSX. Would any reasonable investor say that the portfolio is worth anything more or less than \$2 million?)

If the firm considers investing in a third project C, it should also value C as if C were a mini-firm. That is, the firm should discount the cash flows of C at the expected rate of return that investors would demand if they could make a separate investment in C. *The opportunity cost of capital depends on the use to which that capital is put.*

Perhaps we're saying the obvious. Think of J&J: It is a massive health care and consumer products company, with over \$80 billion in sales. J&J has well-established consumer products, including Band-Aid® bandages, Tylenol®, and products for skin care and babies. It also invests heavily in much chancier ventures, such as biotech research and development (R&D). Do you think that a new production line for baby lotion has the same cost of capital as an investment in biotech R&D? We don't, though we admit that estimating the cost of capital for biotech R&D could be challenging. We will come back to the problems of valuing R&D investments in Chapter 10, when we discuss decision trees, and again in Chapter 23, when we discuss options.

Suppose we measure the risk of each project by its beta. Then J&J should accept any project lying above the upward-sloping security market line that links expected return to risk in Figure 9.1. If the project is high risk, J&J needs a higher prospective return than if the project is low risk. That is not the same as accepting any project *regardless of its risk* as long as it offers a higher return than the *company's* cost of capital. In that case, J&J would accept any project above the horizontal cost of capital line in Figure 9.1—that is, any project offering a return of more than 7.3%.



► FIGURE 9.1

J&J's company cost of capital is about 7.3%. This is the correct discount rate only if the project beta is 0.75. In general, the correct discount rate increases as project beta increases. J&J should accept projects with rates of return above the security market line relating required return to beta.

It wouldn't make sense to suggest that J&J should demand the same rate of return from a very safe project as from a very risky one. If J&J used the company cost of capital rule, it would reject many good low-risk projects and accept many poor high-risk projects. It also wouldn't make sense to suggest that just because another company has a still lower company cost of capital, it is justified in accepting projects that J&J would reject.

Perfect Pitch and the Cost of Capital

The true cost of capital depends on project risk, not on the company undertaking the project. So why is so much time spent estimating the company cost of capital?

There are two reasons. First, many projects can be treated as average risk—that is, neither more nor less risky than the average of the company's other assets. For these projects, the company cost of capital is the right discount rate. Second, the company cost of capital is a useful starting point for setting discount rates for unusually risky or safe projects. It is easier to add to, or subtract from, the company cost of capital than to estimate each project's cost of capital from scratch.

There is a good musical analogy here. Most of us, lacking perfect pitch, need a well-defined reference point, like middle C, before we can sing on key. But anyone who can carry a tune gets *relative* pitches right. Businesspeople have good intuition about *relative* risks, at least in industries they are used to, but not about absolute risk or required rates of return. Therefore, they set a companywide cost of capital as a benchmark. This is not the right discount rate for everything the company does, but adjustments can be made for more or less

risky ventures.

Measuring differences in risk is difficult to do objectively, and financial managers shy away from intracorporate squabbles. (You can imagine the bickering: “My projects are safer than yours! I want a lower discount rate!” “No they’re not! Your projects are riskier than a naked call option!”)⁶ As a result, many companies use the company cost of capital not just as a benchmark, but as an all-purpose discount rate for every project proposal. In this case, they use too low a discount rate for high-risk projects and too high a rate for safer projects. They invest too much in high-risk projects and too little in low-risk projects. Research shows that diversified companies that rely on a single company cost of capital make exactly that mistake and harm shareholders by doing so.⁷

When firms force the use of a single company cost of capital, risk adjustment shifts from the discount rate to project cash flows. Top management may demand extra-conservative cash-flow forecasts from extra-risky projects. Or they may refuse to sign off on an extra-risky project unless NPV, computed at the company cost of capital, is well above zero. Such rough-and-ready risk adjustments may be better than none at all.

9.3 Self-Test

If Johnson & Johnson used the company cost of capital to value all projects, would it accept too many low-risk, or high-risk, projects?

9-2 Estimating Beta and the Company Cost of Capital

Estimating the company cost of capital boils down to estimating the expected return on a portfolio of the company’s equity and debt. Companies with relatively safe debt are usually content to use the promised yield to maturity on the debt as an estimate of its expected return. The harder part is to estimate the expected equity return. To do this, you will probably use the capital asset pricing model (CAPM). If so, you are in good company: Most large U.S. companies use the CAPM to estimate the expected rate of return on their common stock.⁸

We used the CAPM to calculate costs of capital for J&J and CSX in Section 9-1, using estimated equity betas for each company. We did not ask how such betas are estimated in practice. That is our topic now.

Estimating Beta

In principle, we are interested in the future beta of the company's stock, but lacking a crystal ball, we turn first to historical evidence. For example, look at the scatter diagram at the top left of Figure 9.2. Each dot represents the return on U.S. Steel stock and the return on the market in a particular month. The plot starts in March 2010 and runs to February 2015, so there are 60 dots in all.

BEYOND THE PAGE



How to estimate beta

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The second diagram on the left shows a similar plot for the returns on ExxonMobil stock, and the third shows a plot for Consolidated Edison. In each case, we have fitted a line through the points. The slope of this line is an estimate of beta. It tells us how much on average the stock price changed when the market return was 1% higher or lower.

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Try It! Comparing beta estimates

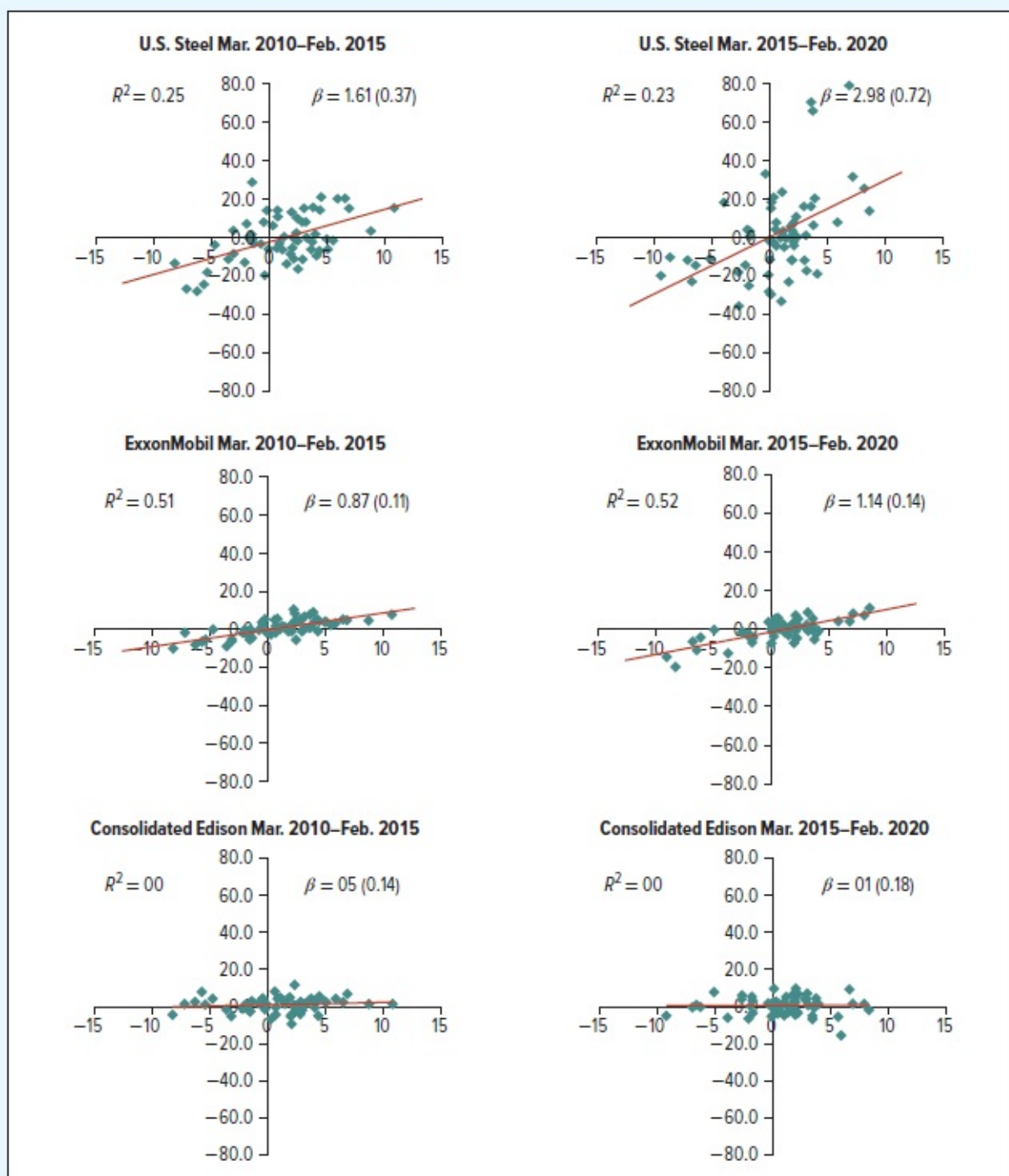
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The right-hand diagrams show similar plots for the same three stocks during the subsequent period ending in February 2020. The estimated betas change. For example, the estimate for U.S. Steel is lower in the first period than in the second. You would have been off target if you had blindly used its beta during the earlier period to predict its beta in the later years. However, you could have been pretty confident that ConEd's beta was much less than U.S. Steel's and that ExxonMobil's beta was somewhere between the two.⁹

Only a portion of each stock's total risk comes from movements in the market. The rest is firm-specific, diversifiable risk, which shows up in the scatter of points around the fitted lines in Figure 9.2. *R-squared* (R^2) measures the proportion of the total variance in the stock's returns that can be explained by market movements. For example, for the first five years the R^2 for ExxonMobil was 0.51. In other words, 51% of ExxonMobil's risk was market risk and 49% was diversifiable risk. The variance of the returns on ExxonMobil stock was 248.¹⁰ So we could say that the variance in stock returns that was due to the market was $0.51 \times 248 = 126$, and the variance of diversifiable returns was $0.49 \times 248 = 122$.

The estimates of beta shown in Figure 9.2 are just that. They are based on the stocks' returns in 60 particular months. The noise in the returns can obscure the true beta. Therefore, statisticians calculate the *standard error* of the estimated beta to show the extent of possible mismeasurement. Then they set up a *confidence interval* of

the estimated value plus or minus two standard errors. We can be much more confident of some estimates than of others. For example, the standard error on ExxonMobil's estimated beta in the second period is 0.14. Thus, the confidence interval for the beta is 1.14 plus or minus 2×0.14 . If you state that the *true* beta for ExxonMobil is between 0.86 and 1.42, you have a 95% chance of being right. We can be much less confident of our estimate of U.S. Steel's beta in the first period. Its standard error is 0.37. So the true beta for U.S. Steel could well be much higher or lower than our estimated figure of 1.61.¹¹



► FIGURE 9.2

We used past returns to estimate the betas of three stocks for the two pre-pandemic periods March 2010 to February 2015 (left-hand diagrams) and March 2015 to February 2020 (right-hand diagrams). In each case the line is fitted by regressing the returns on the stock on returns on the market. Beta is the slope of the fitted line. Notice that in both periods, U.S. Steel had the highest beta and Consolidated Edison the lowest. Standard errors are in parentheses beside the betas. The standard error measures the range of possible error in the beta estimate. We also report the proportion of total risk that is due to market movements (R^2).

Usually, you will have more information (and thus more confidence) than this simple, and somewhat depressing, calculation suggests. For example, you know that ConEd's estimated beta was well below 1 in two successive five-year periods. U.S. Steel's estimated beta was well above 1 in both periods. Nevertheless, there is always a large margin for error when estimating the beta for individual stocks.

Note also that a company's risk may change from period to period. For example, for some companies the COVID-19 pandemic had a dramatic effect on risk. Take Zoom, which had previously catered primarily to corporate clients that used its services for videoconferencing and telecommuting. Zoom was a typical cyclical firm with a business model tied to the general level of business activity; not surprisingly, its beta was above 1, specifically 1.82. Then came the coronavirus and the "work from home" economy. Suddenly, Zoom, which few people had heard of before the pandemic, became an essential tool for both work and socializing. The worse the prognosis for the pandemic (and as a result for the economy), the greater was the demand for Zoom's services. Zoom suddenly became a countercyclical firm, and its beta estimated between February 20, 2020, and June 1, 2020, was actually negative, -0.30 .¹²

By the way, the imprecision in beta estimates cries out for rounding. If someone reports an estimated beta at 1.22664, that someone hasn't looked at the standard error. No one can estimate beta to an accuracy of five decimal places. Savvy practitioners would round to 1.23 or further to 1.2.

Portfolio Betas

Fortunately, the estimation errors tend to cancel out when you estimate betas of *portfolios*.¹³ For example, the beta estimate for a mutual fund with a 50-stock diversified portfolio would have a much lower standard error than the stocks in Figure 9.2. Diversification would remove most of the firm-specific risk that causes the scatter of points around the fitted lines in the figure. It's this scatter that accounts for the imprecision in the beta estimates and the standard errors.

Financial managers sometimes turn to portfolio betas for *industries*. For example, Table

9.1 shows estimates of beta and the standard errors of these estimates for the common stocks of six railroad companies. The standard errors are for the most part over 0.2. CSX's beta, which we used earlier in the chapter, had a standard error of 0.24. However, the estimated beta for a portfolio of all six railroad stocks had a lower standard error of 0.14.

	Beta	Standard Error
Canadian Pacific	1.07	0.18
CSX	1.18	0.24
Kansas City Southern	0.97	0.20
Norfolk Southern	1.33	0.18
Union Pacific	1.09	0.16
Industry portfolio	1.13	0.14

TABLE 9.1 Estimates of betas and standard errors for a sample of railroad companies and for an equally weighted portfolio of these companies, based on monthly returns from March 2015 to February 2020. The portfolio beta may be more reliable than the betas of the individual companies. Note the lower standard error for the portfolio.

EXAMPLE 9.1 • A Railroad Industry Cost of Capital for Berkshire Hathaway

Industry betas are particularly helpful for conglomerate companies investing in many different industries. Berkshire Hathaway is today's largest U.S. conglomerate, with investments in insurance, electric utilities, pipelines, jewelry, chemicals, paints, candies, batteries—the list goes on and on. It also owns BNSF, the Burlington Northern Santa Fe railroad. BNSF is one of the largest U.S. railroads and would have been included in Table 9.1 if it were still an independent public company. BNSF and the other railroads in the table face similar business and operating risks. The cost of capital for the comparable portfolio of railroads should be a good discount rate for Berkshire Hathaway's investments in BNSF.

9.4 Self-Test

Why does diversification increase the accuracy of beta estimates? Explain briefly.

9-3 Analyzing Project Risk

In Section 9.1, we estimated the asset beta for CSX and its company cost of capital. This asset beta is an estimate of the average risk of CSX's railroad business and the company cost of capital is a measure of the expected return on the company as a whole. Not all railroad investments are average risk, however. And if you are the first to use railroad-track networks as deep-space transmission antennas, you will have no asset beta to start with and the company cost of capital will not provide a useful guide to the return that you should demand.

How can you make informed judgments about costs of capital for projects or lines of business when you suspect that risk is *not* average? That is our next topic.

A company that wants to set a cost of capital for one particular line of business typically looks for *pure plays* in that line of business. Pure-play companies are public firms that specialize in one activity. For example, suppose that CSX needs to assess the risk of investing in a new company headquarters. The asset beta for railroads is not helpful. You need to know the beta of commercial real estate. Fortunately, portfolios of commercial real estate are traded. For example, you could estimate asset betas from returns on Real Estate Investment Trusts (REITs) specializing in office buildings.¹⁴ The REITs would serve as Page 258 traded *comparables* for the proposed office building. You could also turn to indexes of real estate prices and returns derived from sales and appraisals of commercial properties.¹⁵

We pointed out earlier that Johnson & Johnson produces a range of consumer products as well as investing in pharmaceuticals and biotech R&D. Suppose that J&J wants to set a cost of capital for its pharmaceutical business. Its company cost of capital could be too low. However, it could estimate the average asset beta or cost of capital for pharmaceutical companies that have *not* diversified into consumer products.

Overall company costs of capital are almost useless for *conglomerates*. Conglomerates diversify into several unrelated industries, so they have to consider industry-specific costs of capital. They therefore look for pure plays in the relevant industries. We have noted Berkshire Hathaway as one example. Take Richard Branson's Virgin Group as another. The group combines many different companies, including airlines (Virgin Atlantic) and train services (Virgin Rail Group). Fortunately, there are examples of pure-play airlines and train operators. The trick is picking the comparables with business risks that are most similar to Virgin's companies.

Sometimes good comparables are not available or are not a good match to a particular project. Then the financial manager has to exercise her or his judgment. Here we offer the following advice:

- *Think about the determinants of asset betas.* Often, the characteristics of high- and low-beta assets can be observed when the beta itself cannot be.
- *Don't be fooled by diversifiable risk.*
- *Avoid fudge factors in discount rates.* Don't give in to the temptation to add fudge factors

to the discount rate to offset things that could go wrong with the proposed investment. Adjust cash-flow forecasts instead.

1. The Determinants of Asset Betas

Cyclical Many people's intuition associates risk with the variability of earnings or cash flow. But much of this variability reflects diversifiable risk. Lone prospectors searching for gold look forward to extremely uncertain future income, but whether they strike it rich is unlikely to depend on the performance of the market portfolio. Even if they do find gold, they do not bear much market risk. Therefore, an investment in gold prospecting has a high standard deviation but a relatively low beta. (Remember from Chapter 7 that Newmont Mining had a high standard deviation, but its beta was close to zero.)

What really counts is the strength of the relationship between the firm's earnings and the aggregate earnings on all real assets. We can measure this either by the *earnings beta* or by the *cash-flow beta*. These are just like a real beta except that changes in earnings or cash flow are used in place of rates of return on securities. Firms with high earnings or cash-flow betas are more likely to have high asset betas.

This means that cyclical firms—firms whose revenues and earnings are strongly dependent on the state of the business cycle—tend to be high-beta firms. Thus, you should demand a higher rate of return from investments whose performance is strongly tied to the performance of the economy.

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Examples of cyclical businesses include airlines, luxury resorts and restaurants, construction, and steel. (Much of the demand for steel depends on construction and capital investment. Note U.S. Steel's sky-high beta in Figure 9.2.) Examples of less-cyclical businesses include food and tobacco products and established consumer brands such as J&J's baby products. A countercyclical example is MBA programs because spending a year or two at a business school is an easier choice when jobs are scarce. Applications to top MBA programs increase in recessions.

9.5 Self-Test

Would you expect a firm of bankruptcy lawyers to have a high or low beta? Explain.

Operating Leverage A production facility with high fixed costs, relative to variable costs, is said to have high *operating leverage*. High operating leverage means a high asset beta. Let us see how this works.

The cash flows generated by an asset can be broken down into revenue, fixed costs, and

variable costs:

$$\text{Cash flow} = \text{revenue} - \text{fixed cost} - \text{variable cost}$$

Costs are variable if they depend on the rate of output. Examples are raw materials, sales commissions, and some labor and maintenance costs. Fixed costs are cash outflows that occur regardless of whether the asset is active or idle, for example, property taxes or the wages of workers under contract. Costs that are fixed do not decline along with sales, and, therefore, any shortfall in sales has a greater impact on profitability.

EXAMPLE 9.2 • Operating Leverage and Risk

The Stamboard Group is comparing two technologies for producing prefabricated housing. Table 9.2 shows the forecasted cash flows. Both technologies generate the same *expected* level of sales and costs, but the costs of Technology A are a constant proportion of sales. By contrast the costs of Technology B are fixed at \$135 million. This higher operating leverage results in a higher asset beta.

Cash flows (\$ millions)			
	Boom	Slump	Expected
Technology A			
Sales	200	100	150
Costs	180	90	135
Net cash flow	20	10	15
Technology B			
Sales	200	100	150
Costs	135	135	135
Net cash flow	65	−35	15

TABLE 9.2 Technologies A and B have the same expected cash flows, but, since B’s costs are fixed, its cash flows are hit harder by economic downturns and have a higher beta.

Now you have a rule of thumb for judging the relative risks of alternative designs or technologies for producing the same project. Other things being equal, the alternative with the higher level of fixed costs will have the higher project beta. Empirical tests confirm that companies with high operating leverage actually do have high betas.¹⁶ We will look more closely at the measurement of operating leverage in the next chapter.

BEYOND THE PAGE



We have interpreted fixed costs as costs of production, but fixed costs can show up in other forms, for example, as future investment outlays. Suppose that an electric utility commits to build a large electricity-generating plant. The plant will take several years to build. During this period the construction costs are fixed obligations. The commitment to invest therefore increases the plant's asset beta. Of course, the commitment disappears when the plant is up and running. Therefore the plant's asset beta is only temporarily high during construction.

Other Sources of Risk So far we have focused on cash flows. Cash-flow risk is not the only risk. A project's value is equal to the expected cash flows discounted at the risk-adjusted discount rate r . If either the risk-free rate or the market risk premium changes, then r will change and so will the project value. We saw in Chapter 3 that prices of long-duration bonds are more exposed to a shift in interest rates than short-duration bonds. In just the same way a project with very long-term cash flows is more exposed to shifts in the discount rate than one with short-term cash flows. This project will, therefore, have a high beta even though it may not have high operating leverage or cyclicalities.¹⁷

You cannot hope to estimate the relative risk of assets with any precision, but good managers examine each project from a variety of angles and look for clues as to its riskiness. They know that high market risk is a characteristic of cyclical ventures, of projects with high fixed costs and of projects that are sensitive to market-wide changes in the discount rate. They think about the major uncertainties affecting the economy and consider how their projects are affected by these uncertainties.

2. Don't Be Fooled by Diversifiable Risk

In this chapter, we have defined risk as the asset beta for a firm, industry, or project. But in everyday usage, "risk" simply means "bad outcome." People think of the risks of a project as a list of things that can go wrong. For example,

A geologist looking for oil worries about the risk of a dry hole.

A pharmaceutical-company scientist worries about the risk that a new drug will have unacceptable side effects.

A plant manager worries that new technology for a production line will fail to work, requiring expensive changes and repairs.

A telecom CFO worries about the risk that a communications satellite will be damaged by space debris. (This was the fate of an Iridium satellite in 2009, when it collided with Russia's defunct Cosmos 2251. Both were blown to smithereens.)

Notice that these risks are all diversifiable. For example, the Iridium-Cosmos collision was definitely a zero-beta event. These hazards do not affect asset betas and should not affect the discount rate for the projects.

Sometimes, financial managers increase discount rates in an attempt to offset these risks. This makes no sense. Diversifiable risks do not increase the cost of capital. They do affect expected cash flows, however.

EXAMPLE 9.3 • Allowing for Possible Bad Outcomes

Project Z will produce just one cash flow, forecasted at \$1 million at year 1. It is regarded as average risk, suitable for discounting at a 10% company cost of capital:

$$PV = \frac{C_1}{1 + r} = \frac{1,000,000}{1.1} = \$909,100$$

But now you discover that the company’s engineers are behind schedule in developing the technology required for the project. They are confident it will work, but they admit to a small chance that it will not. You still see the *most likely* outcome as \$1 million, but you also see some chance that project Z will generate *zero* cash flow.

Now the project’s prospects are clouded by your new worry about technology. It must be worth less than the \$909,100 you calculated before that worry arose. But how much less? There is *some* discount rate (10% plus a fudge factor) that will give the right value, but we do not know what that adjusted discount rate is.

We suggest you reconsider your original \$1 million forecast for project Z’s cash flow. Project cash flows are supposed to be *unbiased* forecasts that give due weight to all possible outcomes, favorable and unfavorable. Managers making unbiased forecasts are correct on average. Sometimes, their forecasts will turn out high, other times low, but their errors will average out over many projects.

If you forecast a cash flow of \$1 million for projects like Z, you will overestimate the average cash flow, because every now and then you will hit a zero. Those zeros should be “averaged in” to your forecasts.

For many projects, the most likely cash flow is also the unbiased forecast. If there are three possible outcomes with the probabilities shown below, the unbiased forecast is \$1 million. (The unbiased forecast is the sum of the probability-weighted cash flows.)

Possible Cash Flow	Probability	Probability-Weighted Cash Flow	Unbiased Forecast
1.2	0.25	0.3	1.0, or \$1 million
1.0	0.50	0.5	
0.8	0.25	0.2	

This might describe the initial prospects of project Z. But if technological uncertainty introduces a 10% chance of a zero cash flow, the unbiased forecast could drop to \$900,000:

Possible Cash Flow	Probability	Probability-Weighted Cash Flow	Unbiased Forecast
1.2	0.225	0.27	0.90, or \$900,000
1.0	0.45	0.45	
0.8	0.225	0.18	
0	0.10	0.0	

The present value is

$$PV = 0.9/1.1 = 0.818, \text{ or } \$818,000$$

Managers often work out a range of possible outcomes for major projects, sometimes with explicit probabilities attached. But even when outcomes and probabilities are not explicitly written down, the manager can still consider the good and bad outcomes as well as the most likely one. When the bad outcomes outweigh the good, the cash-flow forecast should be reduced until balance is regained.

Step 1, then, is to do your best to make unbiased forecasts of a project’s cash flows. Unbiased forecasts incorporate all possible outcomes, including those that are specific to your project and those that stem from economywide events. Step 2 is to consider whether *diversified* investors would regard the project as more or less risky than the average project. In this step only market risks are relevant.

3. Avoid Fudge Factors in Discount Rates

Think back to our example of project Z, where we reduced forecasted cash flows from \$1 million to \$900,000 to account for a possible failure of technology. The project’s PV was reduced from \$909,100 to \$818,000. You could have gotten the right answer by adding a fudge factor to the discount rate and discounting the original forecast of \$1 million at this higher rate. But you have to think through the possible cash flows to get the fudge factor, and once you forecast the cash flows correctly, you don’t need the fudge factor.

Fudge factors in discount rates are dangerous because they displace clear thinking about future cash flows. Here is an example.

EXAMPLE 9.4 • Correcting for Optimistic Forecasts

The chief financial officer (CFO) of EZ² Corp. is disturbed to find that cash-flow forecasts for its investment projects are almost always optimistic. On average they are 10% too high. He therefore decides to compensate by adding 10% to EZ²’s company cost of capital,

increasing it from 12% to 22%.

Suppose the CFO is right about the 10% upward bias in cash-flow forecasts. Can he just add 10% to the discount rate?

Project ZZ has level forecasted cash flows of \$1,000 per year lasting for 15 years. The first two lines of Table 9.3 show these forecasts and their PVs discounted at 12%. Lines 3 and 4 show the corrected forecasts, each reduced by 10%, and the corrected PVs, which are (no surprise) also reduced by 10% (line 5). Line 6 shows the PVs when the uncorrected forecasts are discounted at 22%. The final line 7 shows the percentage reduction in PVs at the 22% discount rate, compared to the unadjusted PVs in line 2.

Year	1	2	3	4	5	...	10	...	15
1. Original cash-flow forecast	\$1,000.00	\$1,000.00	\$1,000.00	\$1,000.00	\$1,000.00	...	\$1,000.00	...	\$1,000.00
2. PV at 12%	\$ 892.90	\$ 797.20	\$ 711.80	\$ 635.50	\$ 567.40	...	\$ 322.00	...	\$ 182.70
3. Corrected cash-flow forecast	\$ 900.00	\$ 900.00	\$ 900.00	\$ 900.00	\$ 900.00	...	\$ 900.00	...	\$ 900.00
4. PV at 12%	\$ 803.60	\$ 717.50	\$ 640.60	\$ 572.00	\$ 510.70	...	\$ 289.80	...	\$ 164.40
5. PV correction	-10.0%	-10.0%	-10.0%	-10.0%	-10.0%	...	-10.0%	...	-10.0%
6. Original forecast discounted at 22%	\$ 819.70	\$ 671.90	\$ 550.70	\$ 451.40	\$ 370.00	...	\$ 136.90	...	\$ 50.70
7. PV "correction" at 22% discount rate	-8.2%	-15.7%	-22.6%	-29.0%	-34.8%	...	-57.5%	...	-72.3%

TABLE 9.3 The original cash-flow forecasts for the ZZ project (line 1) are too optimistic. The forecasts and PVs should be reduced by 10% (lines 3 and 4). But adding a 10% fudge factor to the discount rate reduces PVs by far more than 10% (line 6). The fudge factor overcorrects for bias and would penalize long-lived projects.

Line 5 shows the correct adjustment for optimism (10%). Line 7 shows what happens when a 10% fudge factor is added to the discount rate. The effect on the first year's cash flow is a PV "haircut" of about 8%, 2% less than the CFO expected. But later present values are knocked down by much more than 10%, because the fudge factor is compounded in the 22% discount rate. By years 10 and 15, the PV haircuts are 57% and 72%, far more than the 10% bias that the CFO started with.

Did the CFO really think that bias accumulated as shown in line 7 of Table 9.3? We doubt that he ever asked that question. If he was right in the first place, and the true bias is 10%, then adding a 10% fudge factor to the discount rate understates PV dramatically. The fudge factor also makes long-lived projects look much worse than quick-payback projects.¹⁸

Discount Rates for International Projects

In this chapter, we have concentrated on investments in the United States. In Chapter 28, we say more about investments made internationally. Here, we simply warn against adding fudge factors to discount rates for projects in developing economies. Such fudge factors are too often seen in practice.

It's true that markets are more volatile in developing economies, but much of that risk is diversifiable for investors in the United States, Europe, and other developed countries. It's also true that more things can go wrong for projects in developing economies, particularly in countries that are unstable politically. Expropriations happen. Sometimes governments default on their obligations to international investors. Thus it's especially important to think through the downside risks and to give them weight in cash-flow forecasts.

9.6 Self-Test

The present value of a project has been estimated at \$10 million at a discount rate of 8%, but now the company lawyers point out that there is a 40% chance that the project could result in a costly lawsuit. The CFO therefore increases the discount rate for the project to 9%.

- Is the CFO correct in his approach?
- If the lawsuit would result in costs of \$500,000, all occurring in year 1, what is the PV of the project?

9-4 Certainty Equivalents

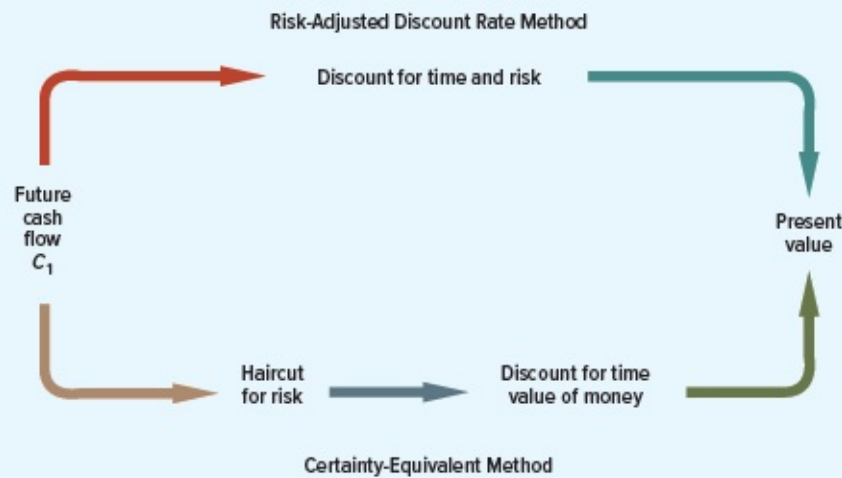
In practical capital budgeting, managers use a risk-adjusted rate to discount future cash flows. This is also the procedure that we have described in this chapter. For example, the discount rate may be calculated from the capital asset pricing model:

$$r = r_f + \beta(r_m - r_f)$$

The resulting r is then substituted directly into the standard discounted cash-flow formula:

$$PV = \sum_{t=1}^T \frac{C_t}{(1+r)^t}$$

The risk-adjusted discount rate, r , adjusts for both time and risk in one step. This is illustrated by the clockwise route in Figure 9.3.



► **FIGURE 9.3**

Two ways to calculate present value. “Haircut for risk” is financial slang referring to the reduction of the cash flow from its forecasted value to its certainty equivalent.

There is an alternative way to value risky cash flows, which makes *separate* adjustments for risk and time. Rather than changing the discount rate, you convert the expected cash flows to **certainty equivalents**, which are then discounted using the risk-free interest rate. This is illustrated by the anti-clockwise route in Figure 9.3. The certainty-equivalent cash flow, or CEQ, is the smallest certain payoff that you would be willing to accept in place of the risky payoff. Since CEQ is a certain payoff, it can be discounted at the risk-free rate.

This gives us two identical expressions for the present value of a cash flow t periods away:¹⁹

$$PV = \frac{C_t}{(1+r)^t} = \frac{CEQ_t}{(1+r_f)^t}$$

Although the certainty-equivalent method is rarely used in everyday project valuation, an understanding of certainty equivalents will allow us to uncover what we are really assuming when we use the same discount rate to value each of a series of future cash flows. It will also become important when we cover options in Chapters 21 through 23 and forward and futures pricing in Chapter 27. Option-pricing formulas discount certainty equivalents. Forward and futures prices *are* certainty equivalents.

BEYOND THE PAGE

**EXAMPLE 9.5 • Certainty Equivalent Valuation**

Think back to the simple real estate investment that we used in Chapter 2 to introduce the concept of present value. You are considering construction of an office building that you plan to sell after one year for \$800,000. That cash flow is uncertain with the same risk as the market, so $\beta = 1$. The risk-free interest rate is $r_f = 7\%$, but you discount the \$800,000 payoff at a risk-adjusted rate of $r = 12\%$. This gives a present value of $800,000/1.12 = \$714,286$.

Suppose a real estate company now approaches and offers to fix the price at which it will buy the building from you at the end of the year. This guarantee would remove any uncertainty about the payoff on your investment. So you would accept a lower figure than the uncertain payoff of \$800,000. But how much less? If the building has a present value of \$714,286 and the interest rate is 7%, then

$$PV = \$714,286 = \frac{\text{certain cash flow}}{1.07}$$
$$\text{Certain cash flow} = \$764,286$$

A project with a guaranteed cash flow of \$764,286 has exactly the same present value as an expected but uncertain cash flow of \$800,000. The cash flow of \$764,286 is therefore known as the *certainty-equivalent cash flow*.

You often hear it said that distant cash flows are more risky than nearby flows and should be discounted at a higher rate. An understanding of certainty equivalents can help to see whether this is correct.

Column 2 in Table 9.4 shows that a project is expected to produce a cash inflow of \$100 million for each of three years. The risk-free interest rate is 6%, the market risk premium is 8% and the project's beta is 0.75. You therefore calculate A's cost of capital as:

$$r = r_f + \beta(r_m - r_f) = 6 + 0.75(8) = 12\%$$

1.	2.	3.	4.	5.
Year	Forecasted Project Cash Flow	PV at 12%	Certainty-Equivalent Cash Flow	Deduction for Risk
1	100	89.3	94.6	5.4
2	100	79.7	89.6	10.4
3	100	71.2	84.8	15.2
		240.2		

TABLE 9.4 This project has a PV of 240.2 when the uncertain cash flows are discounted at a risk-adjusted rate of 12%. It has the same PV when the certainty equivalents are discounted at the risk-free rate of 6%.

Column 3 of Table 9.4 shows the present values of these cash flows at a risk-adjusted discount rate of 12%. Summing these figures gives a total present value for the project of \$240.2.

Column 4 of the table shows the certainty-equivalent cash flows. You can see, for example, that a certain cash flow of 94.6 in year 1 has the same present value as an uncertain cash flow of 100:

$$94.6/1.06 = 100/1.12 = 89.3$$

Since the two cash flows have the same present value, investors must be prepared to give up $100 - 94.6 = 5.4$ of expected income to eliminate uncertainty. Similarly, in year 2 they are prepared to give up $100 - 89.6 = 10.4$ to eliminate uncertainty. This is shown in the final column of Table 9.3, headed “Deduction for Risk.”

When we valued the project, we used the same risk-adjusted discount rate for each year’s cash flow. Now you can see from the final column that this implies a larger deduction for risk from the later cash flows. The reason is that the discount rate compensates for the risk borne *per period*. The more distant the cash flows, the greater the number of periods and the larger the *total* risk adjustment.

9.7 Self-Test

True or false? Assume the project beta is positive.

- Discounting certainty-equivalent cash flows at the project’s cost of capital would double-count risk and understate PV.
- If beta is constant, the ratio of certainty-equivalent to expected cash flows is the same in every year of the project’s life.
- The ratio of certainty-equivalent to expected cash flow is not affected by diversifiable



KEY TAKEAWAYS

In Chapter 8, we set out the basic principles for valuing risky assets. This chapter shows you how to apply those principles when valuing capital investment projects.

- **The company cost of capital (r_A)** is the expected rate of return that investors require on a portfolio of all of the company's outstanding debt and equity. It is the opportunity cost of capital for investment in the firm as a whole. It can be calculated as an average of the expected returns on the firm's equity and debt, weighted by the relative market values of debt and equity:

$$\text{Company cost of capital} = r_A = r_D D/V + r_E E/V$$

- **When should the company cost of capital be used?** Absent taxes, the company cost of capital is the correct discount rate for projects that have the same market risk as the company's existing business. If debt interest is deductible for tax, the company cost of capital must be adjusted for this tax benefit or the value of the tax shield must be added in separately. We explain how to do this in Chapters 17 and 18.
- **How to find the cost of equity (r_E)** Many firms use the CAPM to estimate the cost of equity (r_E). This requires an estimate of the company's beta. Estimates of beta are typically derived from regressions of past stock returns on market returns.
- **Alternative way of finding the company cost of capital (r_A)** Instead of estimating the expected return on both the debt and the equity, managers occasionally estimate the firm's asset beta and substitute this into the CAPM to measure the company cost of capital. The asset beta is a weighted average of the betas of the firm's debt and equity:

$$\beta_A = \beta_D D/V + \beta_E E/V$$

- **When should company cost of capital be used?** Sometimes firms use the same discount rate to value all projects. If this procedure is followed strictly, the firm will accept too many high-risk projects and reject too many low-risk projects. It is *project* risk that counts: The true cost of capital depends on the use to which the capital is put.
- **What determines project risk?** Managers, therefore, need to understand why a particular project may have above- or below-average risk. Cyclical projects and those with high fixed costs generally have high betas.
- **Diversifiable risks and fudge factors** We added two other pieces of advice. Don't be

fooled by diversifiable risk. These do not affect the cost of capital, but the possibility of bad outcomes should be incorporated in the cash-flow forecasts. Also be careful not to offset worries about a project's future performance by adding a fudge factor to the discount rate. Such worries should be reflected in the expected cash flows.

- **Certainty equivalents** Rather than adjusting the discount rate for risk, an alternative is to convert the expected cash flows to certainty equivalents which can then be discounted at the risk-free interest rate. Although seldom used for making capital investment decisions, certainty-equivalents are essential when valuing options. We also used certainty-equivalents to demonstrate how the use of a single risk-adjusted discount rate implies proportionately larger risk deductions for the later cash flows.
- **Finding stock and market risk** The nearby box provides some useful spreadsheet functions for estimating stock and market risk.

USEFUL SPREADSHEET FUNCTIONS

Estimating Stock and Market Risk

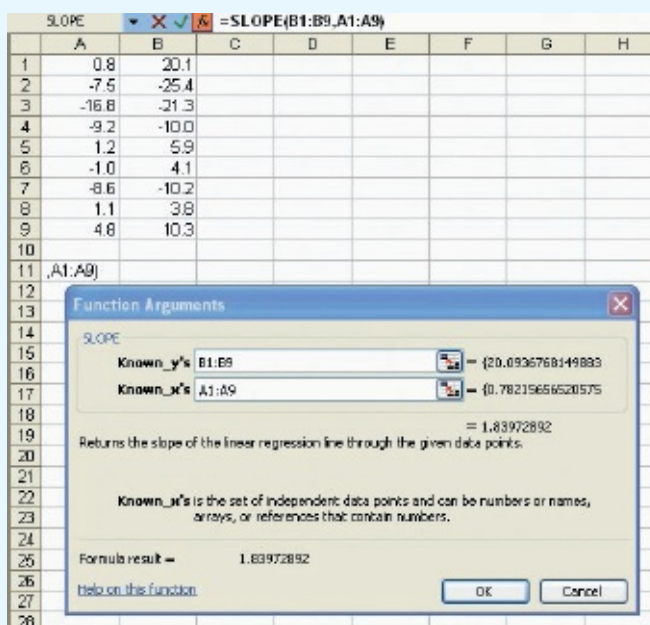
› Spreadsheets such as Excel have some built-in statistical functions that are useful for calculating risk measures. You can find these functions by clicking *fx* on the Excel toolbar. If you then click on the function that you wish to use, Excel will ask you for the inputs that it needs. At the bottom left of the function box, there is a Help facility with an example of how the function is used.

Here is a list of useful functions for estimating stock and market risk. You can enter the inputs for all these functions as numbers or as the addresses of cells that contain the numbers. Note that different versions of Excel may use slightly different names for these functions.

1. **VAR.P** and **STDEV.P**: Calculate variance and standard deviation of a series of numbers, as shown in Section 7-2.
2. **VAR.S** and **STDEV.S**: Footnote 12 of Chapter 7 noted that when variance is estimated from a sample of observations (the usual case), a correction should be made for the loss of a degree of freedom. VAR.S and STDEV.S provide the corrected measures. For any

large sample VAR.S and VAR.P will be similar.

3. **SLOPE:** Useful for calculating the beta of a stock or portfolio.
4. **CORREL:** Useful for calculating the correlation between the returns on any two investments.
5. **COVARIANCE.P** and **COVARIANCE.S:** Portfolio risk depends on the covariance between the returns on each pair of stocks. These functions calculate the covariance.



Microsoft Excel

6. **RSQ:** R-squared is the square of the correlation coefficient and is useful for measuring the proportion of the variance of a stock's returns that can be explained by the market.
7. **AVERAGE:** Calculates the average of any series of numbers.

If, say, you need to know the standard error of your estimate of beta, you can obtain more detailed statistics by going to the *Tools* menu and clicking on *Data Analysis* and then on *Regression*.

Spreadsheet Questions

The following questions provide opportunities to practice each of the Excel functions.

1. (VAR.P and STDEV.P) Choose two well-known stocks and download the latest 61 months of adjusted prices from **finance.yahoo.com**. Calculate the monthly returns for each stock. Now find the variance and standard deviation of the returns for each stock by using VAR.P and STDEV.P. Annualize the variance by multiplying by 12 and the standard deviation by multiplying by the square root of 12.
2. (AVERAGE, VAR.P, and STDEV.P) Now calculate the annualized variance and standard deviation for a portfolio that each month has equal holdings in the two stocks. Is the result more or less than the average of the standard deviations of the two stocks? Why?

3. (SLOPE) Download the Standard & Poor's index for the same period (its symbol is $\hat{GSPC}$). Find the beta of each stock and of the portfolio. (Note: You need to enter the stock returns as the Y-values and market returns as the X-values.) Is the beta of the portfolio more or less than the average of the betas of the two stocks?
4. (CORREL) Calculate the correlation between the returns on the two stocks. Use this measure and your earlier estimates of each stock's variance to calculate the variance of a portfolio that is evenly divided between the two stocks. (You may need to reread Section 7-3 to refresh your memory of how to do this.) Check that you get the same answer as when you calculated the portfolio variance directly.
5. (COVARIANCE.P) Repeat Question 4, but now calculate the covariance directly rather than from the correlations and variances.
6. (RSQ) For each of the two stocks calculate the proportion of the variance explained by the market index. Do the results square with your intuition?
7. Use the *Regression* facility under the *Data Analysis* menu to calculate the beta of each stock and of the portfolio (beta here is called the coefficient of the X-variable). Look at the standard error of the estimate in the cell to the right. How confident can you be of your estimates of the betas of each stock? How about your estimate of the portfolio beta?

FURTHER READING

For a useful survey of the issues covered in this chapter, see:

R. Jagannathan, J. Liberti, B. Liu, and I. Meier, "A Firm's Cost of Capital," *Annual Review of Financial Economics* 9 (November 2017), pp. 259–282.

Levi and Welch provide some recommendations for estimating betas in:

Y. Levi and I. Welch, "Best Practice for Cost-of-Capital Estimates," *Journal of Financial and Quantitative Analysis* 52 (April 2017), pp. 427–463.

PROBLEM SETS



connect

Select problems are available in McGraw-Hill's *Connect*.

Please see the preface for more information.

1. **Definitions (S9.1–S9.3)** Define the following terms:

- a. Cost of debt.
- b. Cost of equity.
- c. Company cost of capital.
- d. Equity beta.
- e. Asset beta.
- f. Pure-play comparable.

2. True/false (S9.1–S9.3) True or false?

- a. The company cost of capital is the correct discount rate for all projects because the high risks of some projects are offset by the low risk of other projects.
- b. Adding fudge factors to discount rates undervalues long-lived projects compared with quick-payoff projects.

3. Company cost of capital (S9.1) Quark Productions (“Give your loved one a quark today.”) uses its company cost of capital to evaluate all projects. Will it underestimate or overestimate the value of high-risk projects?

4. Company cost of capital (S9.2) The total market value of the common stock of the Okefenokee Real Estate Company is \$6 million, and the total value of its debt is \$4 million. The treasurer estimates that the beta of the stock is currently 1.5 and that the expected risk premium on the market is 6%. The Treasury bill rate is 4%. Assume for simplicity that Okefenokee debt is risk-free and the company does not pay tax.

- a. What is the required return on Okefenokee stock?
- b. Estimate the company cost of capital.
- c. What is the discount rate for an expansion of the company’s present business?
- d. Suppose the company wants to diversify into the manufacture of rose-colored spectacles. The beta of optical manufacturers that have no debt is 1.2. Estimate the required return on Okefenokee’s new venture.

5. Company cost of capital (S9.2) You are given the following information Page 269
for Golden Fleece Financial:

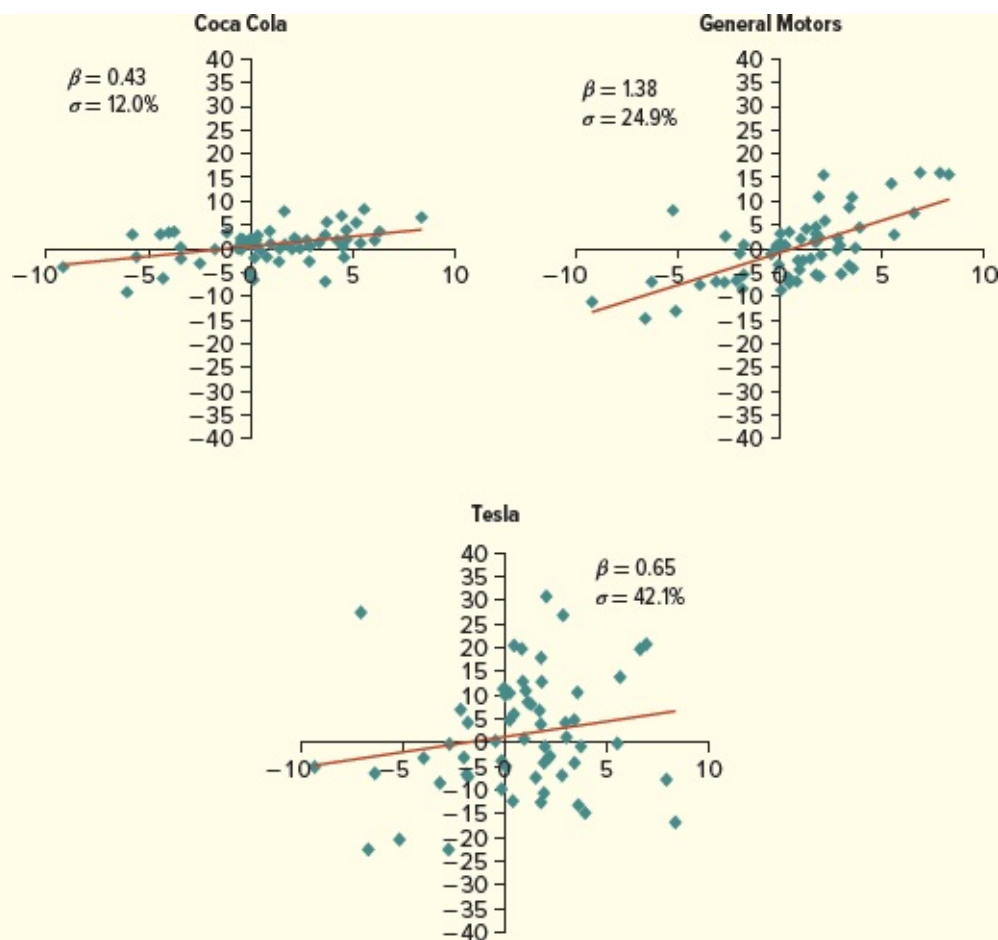
Long-term debt outstanding:	\$300,000
Current yield to maturity (r_{debt}):	8%
Number of shares of common stock:	10,000
Price per share:	\$50
Book value per share:	\$25
Expected rate of return on stock (r_{equity}):	15%

Calculate Golden Fleece’s company cost of capital.

6. Company cost of capital (S9.2) Nero Violins has the following capital structure:

Security	Beta	Total Market Value (\$ millions)
Debt	0	\$100
Preferred stock	0.20	40
Common stock	1.20	299

- a. What is the firm's asset beta? (*Hint: What is the beta of a portfolio of all the firm's securities?*)
 - b. Assume that the CAPM is correct. What discount rate should Nero set for investments that expand the scale of its operations without changing its asset beta? Assume a risk-free interest rate of 5% and a market risk premium of 6%. Ignore taxes.
7. **Company cost of capital (S9.2)** A company is 40% financed by risk-free debt. The interest rate is 10%, the expected market risk premium is 8%, and the beta of the company's common stock is 0.5. What is the company cost of capital?
 8. **Company cost of capital (S9.2)** Binomial Tree Farm's financing includes \$5 million of bank loans. Its common equity is shown in Binomial's Annual Report at \$6.67 million. It has 500,000 shares of common stock outstanding, which trade on the Wichita Stock Exchange at \$18 per share. What debt ratio should Binomial use to calculate its company cost of capital or asset beta? Explain.
 9. **Measuring risk (S9.2)** Refer to the top-right panel of Figure 9.2. What proportion of U.S. Steel's returns was explained by market movements? What proportion of risk was diversifiable? How does the diversifiable risk show up in the plot? What is the range of possible errors in the estimated beta?
 10. **Measuring risk (S9.2)** Figure 9.4 shows plots of monthly rates of return on three stocks versus those of the market index. The beta and standard deviation of each stock is given beside the plot.
 - a. Which stock is safest for a diversified investor?
 - b. Which stock is safest for an undiversified investor who puts all her money in one of these stocks?
 - c. Consider a portfolio with equal investments in each stock. What would be this portfolio's beta?
 - d. Consider a well-diversified portfolio composed of stocks with the same beta and standard deviation as Tesla. What are the beta and standard deviation of this portfolio's return? The standard deviation of the market portfolio's return is 20%.
 - e. Use the capital asset pricing model to estimate the expected return on each stock. The risk-free rate is 4%, and the market risk premium is 8%.



► **FIGURE 9.4**

- 11. Measuring risk (S9.2)** The following table shows estimates of the risk of two well-known Canadian stocks:

	Standard Deviation (%)	R^2	Beta	Standard Error of Beta
Sun Life Financial	18.3	0.34	0.89	0.16
Suncor Energy	23.4	0.32	1.11	0.21

- What proportion of each stock's risk was market risk, and what proportion was specific risk?
- What is the variance of the returns for Sun Life Financial stock? What is the specific variance?
- What is the confidence interval on Suncor's beta? (See Section 9-2 on Estimating Beta for a definition of "confidence interval.")
- If the CAPM is correct, what is the expected return on Sun Life? Assume a risk-free interest rate of 5% and an expected market return of 12%.
- Suppose that next year, the market provides a 20% return. Knowing this, what return would you expect from Sun Life?

- 12. Measuring risk (S9.1)** Look again at Table 9.1. This time we will concentrate on Union Pacific.
- Calculate Union Pacific's cost of equity from the CAPM using its own beta estimate and the industry beta estimate. How different are your answers? Assume a risk-free rate of 2% and a market risk premium of 7%.
 - Can you be confident that Union Pacific's true beta is not the industry average?
 - Under what circumstances might you advise Union Pacific to calculate its cost of equity based on its own beta estimate?
 - You now discover that the estimated beta for Union Pacific in the previous five-year period was 1.3. Does this influence your answer to part (c)?
- 13. Asset betas (S9.2)** EZCUBE Corp. is 50% financed with long-term bonds and 50% with common equity. The debt securities have a beta of .15. The company's equity beta is 1.25. What is EZCUBE's asset beta?
- 14. Asset betas (S9.3)** Which of these projects is likely to have the higher asset beta, other things equal? Why?
- The sales force for project A is paid a fixed annual salary. Project B's sales force is paid by commissions only.
 - Project C is a first-class-only airline. Project D is a well-established line of breakfast cereals.
- 15. Asset betas (S9.3)** What types of firms need to estimate industry asset betas? How would such a firm make the estimate? Describe the process step by step.
- 16. Betas and operating leverage (S9.3)** You run a perpetual encabulator machine, which generates revenues averaging \$20 million per year. Raw material costs are 50% of revenues. These costs are variable—they are always proportional to revenues. There are no other operating costs. The cost of capital is 9%. Your firm's long-term borrowing rate is 6%. Now you are approached by Studebaker Capital Corp., which proposes a fixed-price contract to supply raw materials at \$10 million per year for 10 years.
- What happens to the operating leverage and business risk of the encabulator machine if you agree to this fixed-price contract?
 - Calculate the present value of the encabulator machine with and without the fixed-price contract.
- 17. Diversifiable risk (S9.3)** Many investment projects are exposed to diversifiable risks. What does "diversifiable" mean in this context? How should diversifiable risks be accounted for in project valuation? Should they be ignored completely?
- 18. Fudge factors (S9.3)** Mario Barleycorn estimates his firm's company cost of capital at only 8%. Nevertheless, he sets a 15% companywide discount rate to offset the optimistic biases of project sponsors and to impose "discipline" on the capital budgeting process. Suppose Mr. Barleycorn is correct about the project sponsors,

who are, in fact, optimistic by 7% on average. Explain why the increase in the discount rate from 8% to 15% will not offset the bias. (Note: The firm is entirely equity financed.)

19. Fudge factors (S9.3) Mom and Pop Groceries has just dispatched a year's supply of groceries to the government of the Central Antarctic Republic. Payment of \$250,000 will be made one year hence after the shipment arrives by snow train. Unfortunately, there is a good chance of a coup d'état, in which case the new government will not pay. Mom and Pop's controller therefore decides to discount the payment at 40% rather than at the company's 12% cost of capital.

- a. What's wrong with using a 40% rate to offset political risk?
- b. How much is the \$250,000 payment really worth if the odds of a coup d'état are 25%?

20. Fudge factors (S9.3) An oil company is drilling a series of new wells on the perimeter of a producing oil field. About 20% of the new wells will be dry holes. Even if a new well strikes oil, there is still uncertainty about the amount of Page 272 oil produced: 40% of new wells that strike oil produce only 1,000 barrels a day; 60% produce 5,000 barrels per day.

- a. Forecast the annual cash revenues from a new perimeter well. Use a future oil price of \$100 per barrel.
- b. A geologist proposes to discount the cash flows of the new wells at 30% to offset the risk of dry holes. The oil company's normal cost of capital is 10%. Does this proposal make sense? Briefly explain why or why not.

21. Certainty equivalents (S9.4) A project has a forecasted cash flow of \$110 in year 1 and \$121 in year 2. The interest rate is 5%, the estimated risk premium on the market is 10%, and the project has a beta of 0.5. If you use a constant risk-adjusted discount rate, what is

- a. The PV of the project?
- b. The certainty-equivalent cash flow in years 1 and 2?
- c. The ratio of the certainty-equivalent cash flows to the expected cash flow?

22. Certainty equivalents (S9.4) A project has the following forecasted cash flows:

Cash Flows (\$ thousands)			
C_0	C_1	C_2	C_3
-100	+40	+60	+50

The estimated beta is 1.5. The market return is 16%, and the risk-free rate is 7%.

- a. Estimate the cost of capital for the project and the project's PV.
- b. What are the certainty-equivalent cash flows in each year?

- c. What is the ratio of the certainty-equivalent cash flow to the expected cash flow in each year?
- d. Explain why this ratio declines.

CHALLENGE

- 23. Beta of costs (S9.2)** Suppose that you are valuing a future stream of high-risk (high-beta) cash *outflows*. High risk means a high discount rate. But the higher the discount rate, the less the present value. This seems to say that the higher the risk of cash outflows, the less you should worry about them! Can that be right? Should the sign of the cash flow affect the appropriate discount rate? Explain.
- 24. Fudge factors (S9.3)** An oil company executive is considering investing \$10 million in one or both of two wells: Well 1 is expected to produce oil worth \$3 million a year for 10 years; well 2 is expected to produce \$2 million for 15 years. These are real (inflation-adjusted) cash flows.

The beta for producing wells is 0.9. The market risk premium is 8%, the nominal risk-free interest rate is 6%, and expected inflation is 4%.

The two wells are intended to develop a previously discovered oil field. Unfortunately there is still a 20% chance of a dry hole in each case. A dry hole means zero cash flows and a complete loss of the \$10 million investment.

Ignore taxes and make further assumptions as necessary.

- a. What is the correct real discount rate for cash flows from developed wells?
- b. The oil company executive proposes to add 20 percentage points to the real discount rate to offset the risk of a dry hole. Calculate the NPV of each well with this adjusted discount rate.
- c. What do you say the true NPVs of the two wells are?
- d. Is there any single fudge factor that could be added to the discount rate for developed wells that would yield the correct NPV for both wells? Explain.



SOLUTIONS TO SELF-TEST QUESTIONS

- 9.1** Too low. The market value of equity is generally well above book value, and the expected return on equity is always higher than that on debt.
- 9.2** $D/V = 0.33$ and $E/V = 0.67$. $r_D = 4\%$ and $r_E = 2 + 1 \times 7 = 9\%$. The company cost of

capital, therefore, equals $r_A = r_D D/V + r_E E/V = 4 \times 0.33 + 9 \times 0.67 = 7.33\%$.

9.3 Too many high-risk projects, too few low-risk projects.

9.4 Diversification reduces portfolio-specific risks, which reduces the scatter of returns around the line relating portfolio and market returns. This makes it easier to locate the line and measure its slope, which is beta.

9.5 Bankruptcy lawyers are likely to have more business in recessions. Their firms may even have negative betas.

9.6

- a. No, the chance of a lawsuit is unlikely to be related to the state of the economy and is not a market risk.
- b. PV should be reduced by $(0.4 \times 500,000)/1.08 = \$185,000$.

9.7

- a. True
- b. False
- c. True



FINANCE ON THE WEB

You can download data for the following questions from finance.yahoo.com or any site that provides financial data.

1. Look at the companies listed in Table 8.4. Calculate monthly rates of return for two successive five-year periods. Calculate betas for each subperiod using the Excel SLOPE function. How stable was each company's beta? Suppose that you had used these betas to estimate expected rates of return from the CAPM. Would your estimates have changed significantly from period to period?
2. Identify a sample of food companies. For example, you could try Campbell Soup (CPB), General Mills (GIS), Kellogg (K), Mondelez International (MDLZ), and Tyson Foods (TSN).
 - a. Estimate beta and R^2 for each company, using five years of monthly returns and Excel functions SLOPE and RSQ.
 - b. Average the returns for each month to give the return on an equally weighted portfolio of the stocks. Then calculate the industry beta using these portfolio

returns. How does the R^2 of this portfolio compare with the average R^2 of the individual stocks?

- c. Use the CAPM to calculate an average cost of equity (r_{equity}) for the food industry. Use current interest rates—take a look at the discussion of what is the risk-free rate? in Section 9-1—and a reasonable estimate of the market risk premium.

MINI-CASE

The Jones Family Incorporated

The Scene: It is early evening in the summer of 2021, in an ordinary family room in Manhattan. Modern furniture, with old copies of *The Wall Street Journal* and the *Financial Times* scattered around. Autographed photos of Jerome Powell and George Soros are prominently displayed. A picture window reveals a distant view of lights on the Hudson River. John Jones sits at a computer terminal, glumly sipping a glass of chardonnay and putting on a carry trade in Japanese yen over the Internet. His wife Marsha enters.

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Marsha: Hi, honey. Glad to be home. Lousy day on the trading floor, though. Dullsville. No volume. But I did manage to hedge next year's production from our copper mine. I couldn't get a good quote on the right package of futures contracts, so I arranged a commodity swap.
John doesn't reply.

Marsha: John, what's wrong? Have you been selling yen again? That's been a losing trade for weeks.

John: Well, yes. I shouldn't have gone to Goldman Sachs's foreign exchange brunch. But I've got to get out of the house somehow. I'm cooped up here all day calculating covariances and efficient risk-return trade-offs while you're out trading commodity futures. You get all the glamour and excitement.

Marsha: Don't worry, dear, it will be over soon. We only recalculate our most efficient common stock portfolio once a quarter. Then you can go back to leveraged leases.

John: You trade, and I do all the worrying. Now there's a rumor that our leasing company is going to get a hostile takeover bid. I knew the debt ratio was too low, and you forgot to put on the poison pill. And now you've made a negative-NPV investment!

Marsha: What investment?

John: That wildcat oil well. Another well in that old Sourdough field. It's going to cost \$5 million! Is there any oil down there?

Marsha: That Sourdough field has been good to us, John. Where do you think we got the

capital for your yen trades? I bet we'll find oil. Our geologists say there's only a 30% chance of a dry hole.

ohn: Even if we hit oil, I bet we'll only get 75 barrels of crude oil per day.

larsha: That's 75 barrels day in, day out. There are 365 days in a year, dear.

ohn and Marsha's teenage son Johnny bursts into the room.

ohnny: Hi, Dad! Hi, Mom! Guess what? I've made the junior varsity derivatives team! That means I can go on the field trip to the Chicago Board Options Exchange. (*Pauses.*) What's wrong?

ohn: Your mother has made another negative-NPV investment. A wildcat oil well, way up on the North Slope of Alaska.

ohnny: That's OK, Dad. Mom told me about it. I was going to do an NPV calculation yesterday, but I had to finish calculating the junk-bond default probabilities for my corporate finance homework. (*Grabs a financial calculator from his backpack.*) Let's see: 75 barrels a day times 365 days per year times \$100 per barrel when delivered in Los Angeles ... that's \$2.7 million per year.

ohn: That's \$2.7 million *next* year, assuming that we find any oil at all. The production will start declining by 5% every year. And we still have to pay \$20 per barrel in pipeline and tanker charges to ship the oil from the North Slope to Los Angeles. We've got some serious operating leverage here.

larsha: On the other hand, our energy consultants project increasing oil prices. If they increase with inflation, price per barrel should increase by roughly 2.5% per year. The wells ought to be able to keep pumping for at least 15 years.

ohnny: I'll calculate NPV after I finish with the default probabilities. The interest rate is 6%. Is it OK if I work with the beta of 0.8 and our usual figure of 7% for the market risk premium?

larsha: I guess so, Johnny. But I am concerned about the fixed shipping costs.

ohn: (*Takes a deep breath and stands up.*) Anyway, how about a nice family dinner? I've reserved our usual table at the Four Seasons.

Everyone exits.

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announcer: Is the wildcat well really negative NPV? Will John and Marsha have to fight a hostile takeover? Will Johnny's derivatives team use Black-Scholes or the binomial method? Find out in the next episode of The Jones Family Incorporated.

You may not aspire to the Jones family's way of life, but you will learn about all their activities, from futures contracts to binomial option pricing, later in this book. Meanwhile, you may wish to replicate Johnny's NPV analysis.

QUESTIONS

1. Calculate the NPV of the wildcat oil well, taking account of the probability of a dry hole, the shipping costs, the decline in production, and the forecasted increase in oil prices.

How long does production have to continue for the well to be a positive-NPV investment? Ignore taxes and other possible complications.

2. Now consider operating leverage. How should the shipping costs be valued, assuming that output is known and the costs are fixed? How would your answer change if the shipping costs were proportional to output? Assume that unexpected fluctuations in output are zero-beta and diversifiable. (*Hint: The Jones's oil company has an excellent credit rating. Its long-term borrowing rate is only 7%.*)

¹J&J's market cap at the end of the third quarter of 2020 was about \$380 billion, its long-term debt about \$30 billion. It also had \$12 billion invested in marketable debt securities, so its net debt was only \$18 billion.

²Sometimes practitioners assume without thinking or checking that $\beta_D = 0$. This can be a decent approximation for the most creditworthy corporations when interest rates are stable. The approximation breaks down as the spread between treasury and corporate interest rates increases. The spread reflects several issues, including default risk and the illiquidity of corporate bonds.

³It is identical because the 3.4% cost of debt is consistent with a debt beta of $\beta_D = 0.2$. The two calculations diverge if the observed cost of debt differs from the cost of debt that would be calculated from the CAPM.

⁴The other muddle would use a forecast of future short rates in the CAPM formula. Making such a forecast in late 2020, with interest rates at record lows at the height of the COVID-19 pandemic, was foolhardy at best. If a long-term forecast of short rates was needed, it was probably better to use some average, "normal" rate, say 2%, as in our examples above.

⁵This approach gives a security market line with a higher intercept and a lower market risk premium. Using a "flatter" security market line is perhaps a better match to the historical evidence, which shows that the slope of average returns against beta is not as steeply upward-sloping as the CAPM predicts. See Figure 8.5.

⁶A "naked" call option is an option purchased with no offsetting (hedging) position in the underlying stock or in other options. We discuss options in Chapter 21.

⁷P. Krüger, A. Landier, and D. Thesmar, "The WACC Fallacy: The Real Effects of Using a Unique Discount Rate," *Journal of Finance* 70 (June 2015), pp. 1253–1285.

⁸The CAPM is not the last word on risk and return, of course, but the principles and procedures covered in this chapter work just as well with other models such as the Fama–French three-factor model. See Section 8-4.

⁹Remember that to estimate beta, you must regress the *returns* on the stock (the *y* variable) on the market *returns* (the *x* variable). You would get a very similar estimate if you simply used the percentage *changes* in the stock price and the market index. But sometimes, people make the mistake of regressing the stock price *level* on the *level* of the index and obtain nonsense results.

¹⁰This is an annual figure; we annualized the monthly variance by multiplying by 12 (see footnote 14 in Chapter 7). The standard deviation was $\sqrt{248} = 15.8\%$.

¹¹For this reason, statisticians would generally adjust this estimate of U.S. Steel's beta down toward 1.0. Similarly, they would adjust upward any estimated beta that was below 1.0. For simplicity, in this book we use unadjusted betas, but if you are interested in how to make the adjustment, take a look at the Beyond the Page feature, "How to estimate beta."

¹²See Karen Langley, "For Some Stocks, Bad News Is Now Good News," *The Wall Street Journal*, July 20, 2020.

¹³If the observations are independent, the standard error of the estimated mean beta declines in proportion to the square root of the number of stocks in the portfolio.

¹⁴REITs are investment funds that invest in real estate. You would have to be careful to identify REITs investing in commercial properties similar to the proposed office building. There are also REITs that invest in other types of real estate, including apartment buildings, shopping centers, and timberland.

¹⁵See Chapter 23 in D. Geltner, N. G. Miller, J. Clayton, and P. Eichholtz, *Commercial Real Estate Analysis and*

Investments, 3rd ed. (South-Western College Publishing, 2014).

¹⁶See B. Lev, "On the Association between Operating Leverage and Risk," *Journal of Financial and Quantitative Analysis* 9 (September 1974), pp. 627–642; and G. N. Mandelker and S. G. Rhee, "The Impact of the Degrees of Operating and Financial Leverage on Systematic Risk of Common Stock," *Journal of Financial and Quantitative Analysis* 19 (March 1984), pp. 45–57.

¹⁷See J. Y. Campbell and J. Mei, "Where Do Betas Come From? Asset Price Dynamics and the Sources of Systematic Risk," *Review of Financial Studies* 6 (Fall 1993), pp. 567–592. Cornell discusses the effect of duration on project risk in B. Cornell, "Risk, Duration and Capital Budgeting: New Evidence on Some Old Questions," *Journal of Business* 72 (April 1999), pp. 183–200.

¹⁸The optimistic bias could be worse for distant than near cash flows. If so, the CFO should make the time-pattern of bias explicit and adjust the cash-flow forecasts accordingly.

¹⁹CEQ1 can be calculated directly from the capital asset pricing model. The certainty-equivalent value of the cash flow C_1 is $C_1 - \lambda \text{cov}(\tilde{C}_1, r_m)$. $\text{Cov}(\tilde{C}_1, r_m)$ is the covariance between the uncertain cash flow and the return on the market. Lambda, λ , is a measure of the market price of risk. It is defined as $(r_m - r_f)/\sigma_m^2$. For example, if $r_m - r_f = 0.08$ and the standard deviation of the market is $\sigma_m = 0.20$, then $\lambda = 0.08/0.20^2 = 2$.